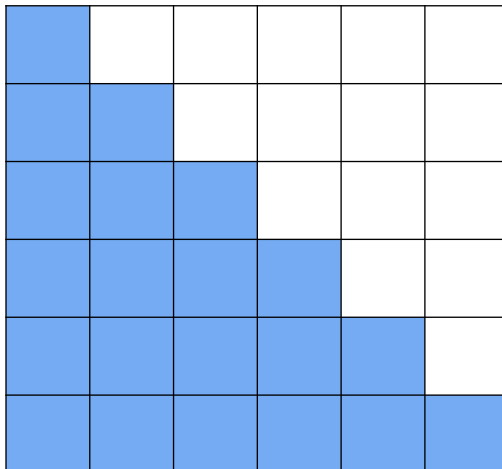

Sparse Attention

INTRODUCTION TO LLM
INFERENCE SERVING SYSTEMS

CHUHONG YUAN

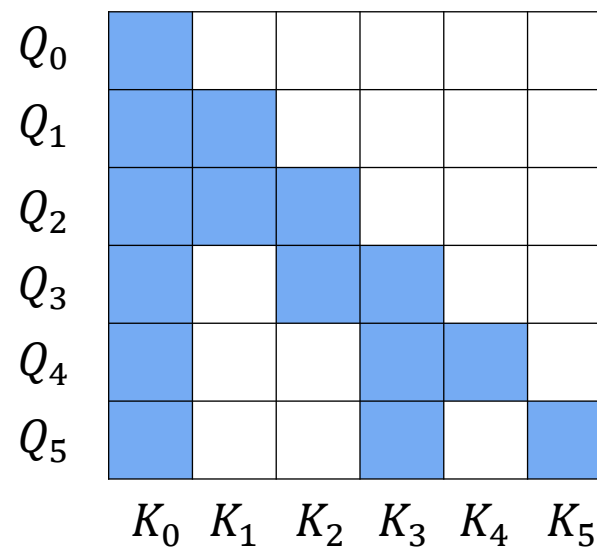
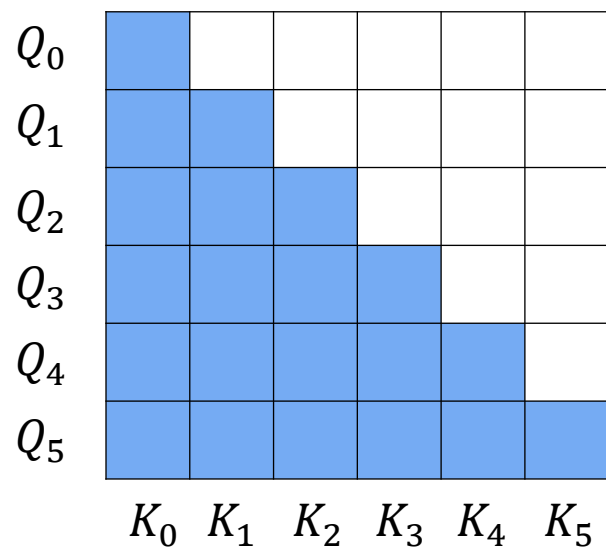


Dense vs. Sparse Attention

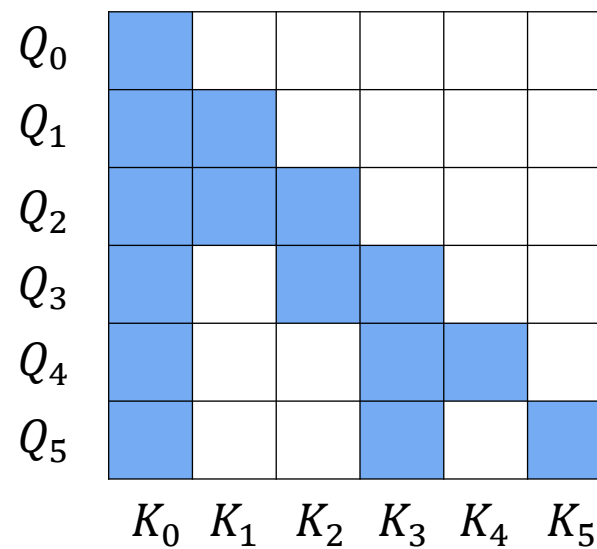
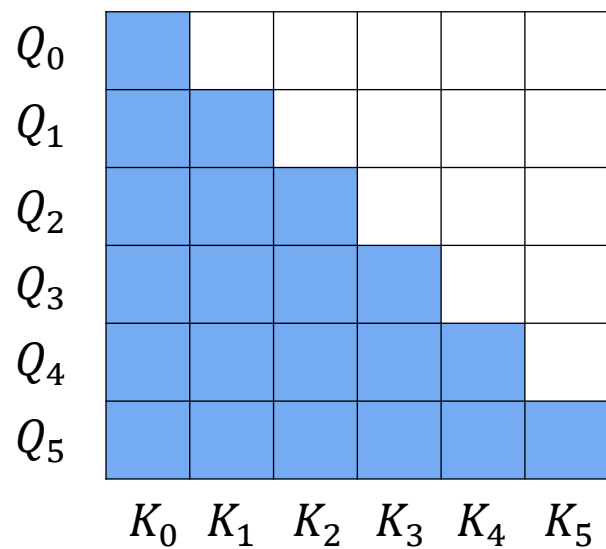


Q_0						
Q_1						
Q_2						
Q_3						
Q_4						
Q_5						
	K_0	K_1	K_2	K_3	K_4	K_5

Dense vs. Sparse Attention



Dense vs. Sparse Attention



Use a mask matrix



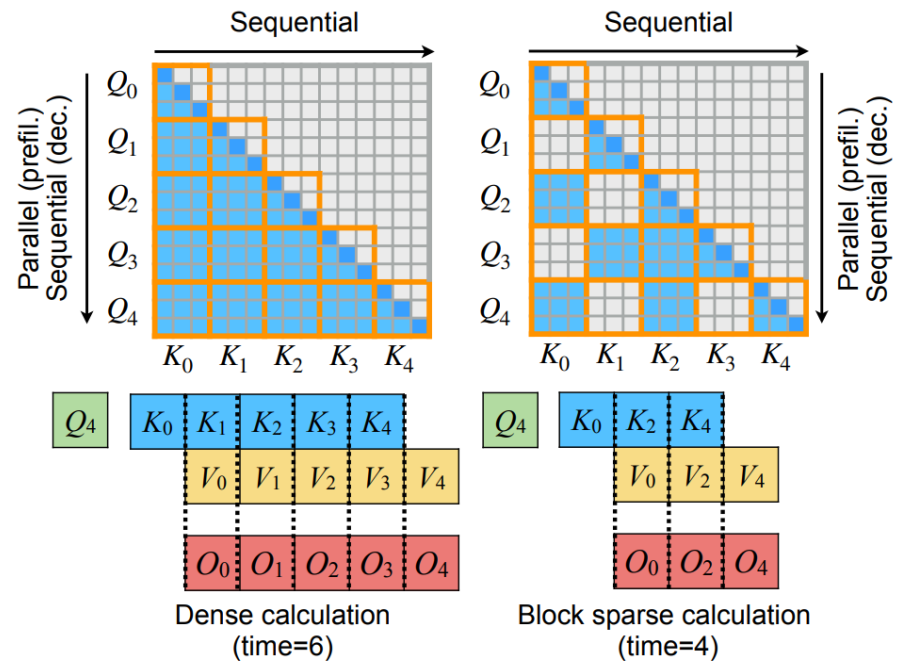
Sparse Attention Motivation

- Sparsity of attention scores
- KV Cache of different sequences
- Long-context requests handling
- Speculative decoding

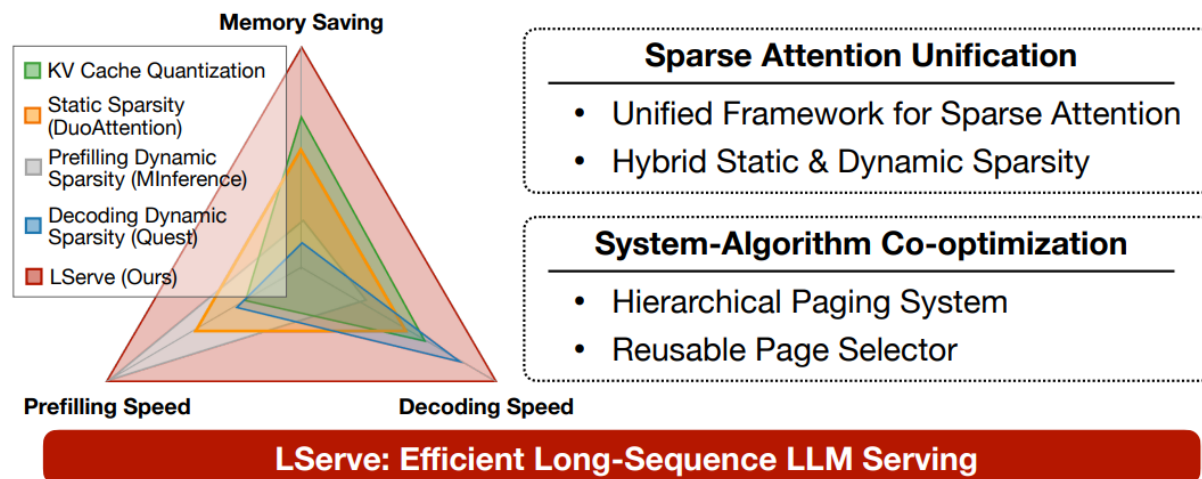
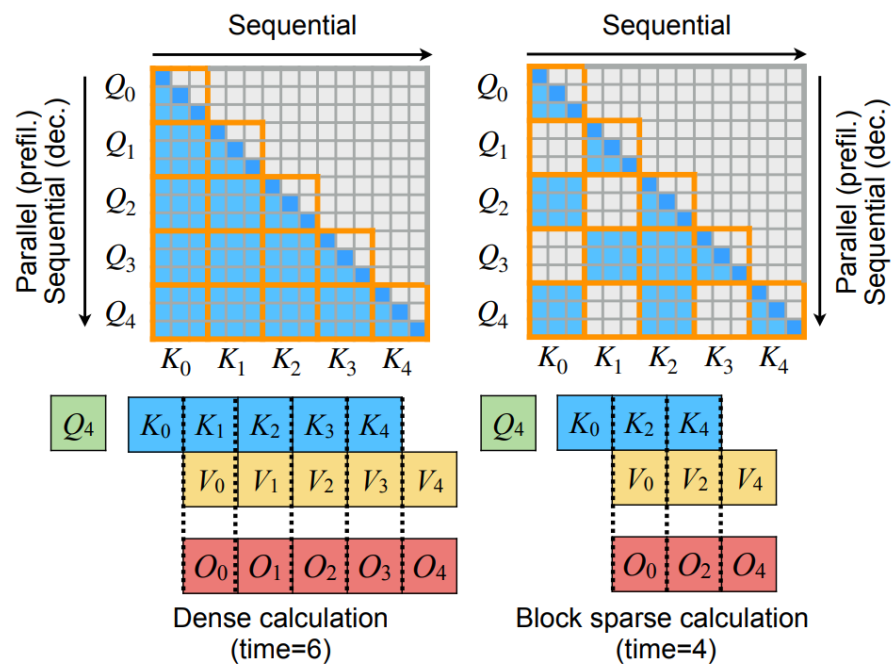
LServe



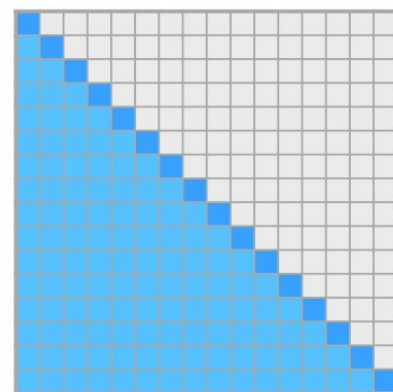
Motivation



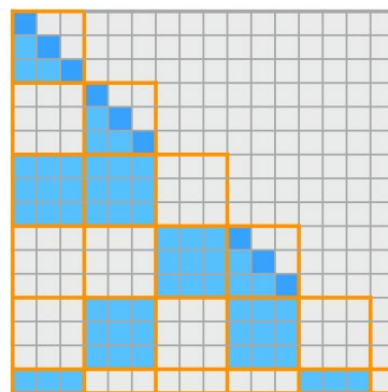
Motivation



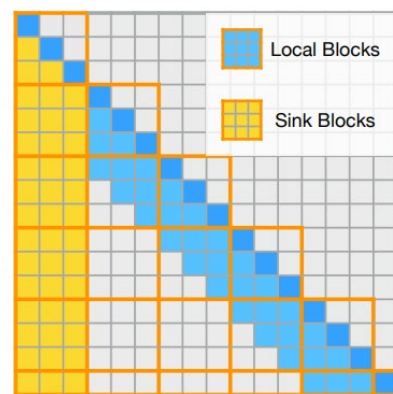
Unified Block Sparse Attention



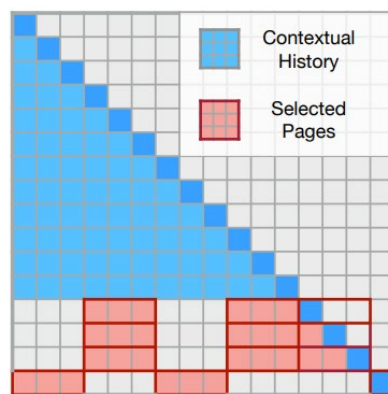
(a) Dense Attention



(b) Block-Sparse Attention

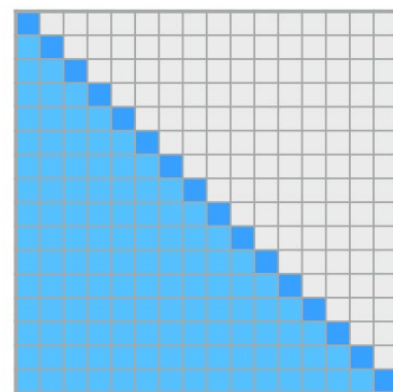


(c) Block-Sparse: Streaming Heads

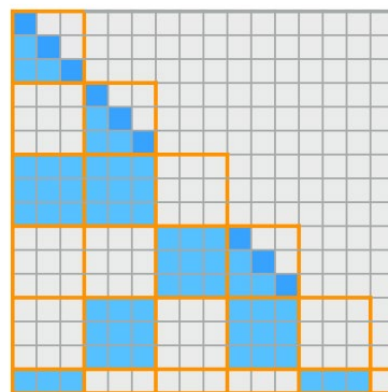


(d) Block-Sparse: Page Pruning

Unified Block Sparse Attention

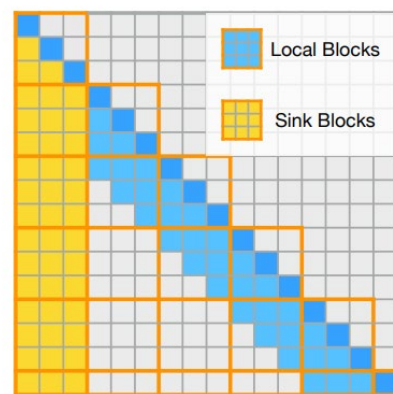


(a) Dense Attention

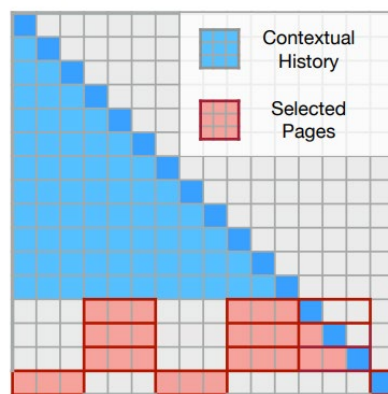


(b) Block-Sparse Attention

Sparsity defined at the block level

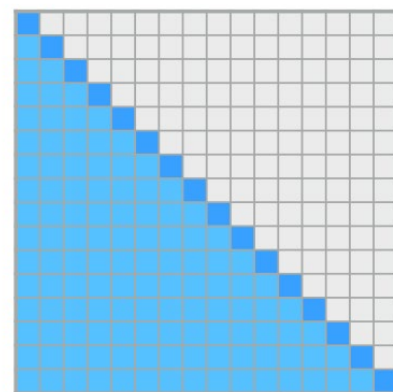


(c) Block-Sparse: Streaming Heads

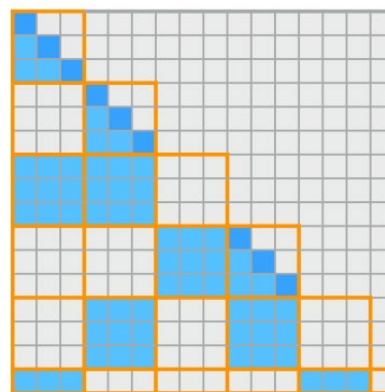


(d) Block-Sparse: Page Pruning

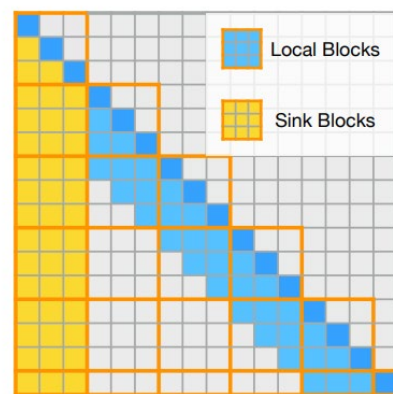
Unified Block Sparse Attention



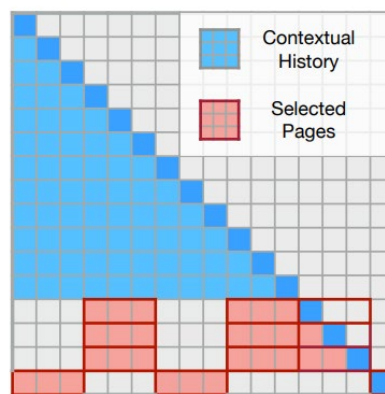
(a) Dense Attention



(b) Block-Sparse Attention



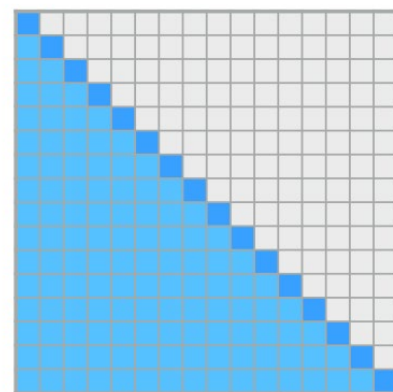
(c) Block-Sparse: Streaming Heads



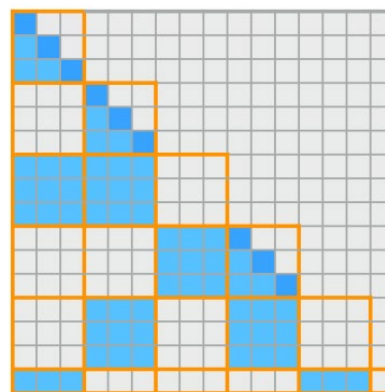
(d) Block-Sparse: Page Pruning

Offline and static

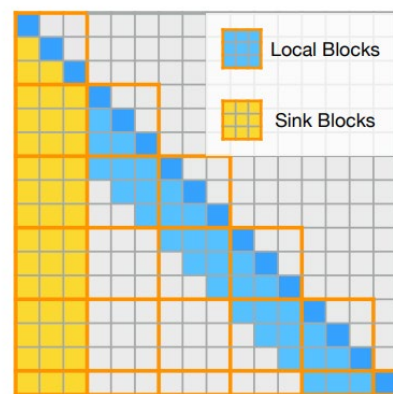
Unified Block Sparse Attention



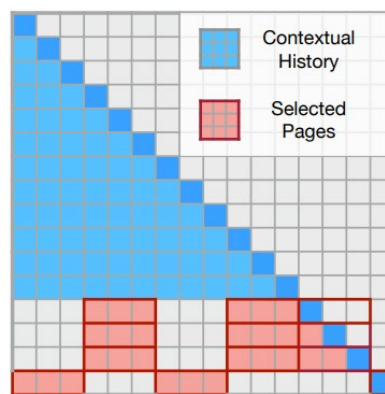
(a) Dense Attention



(b) Block-Sparse Attention



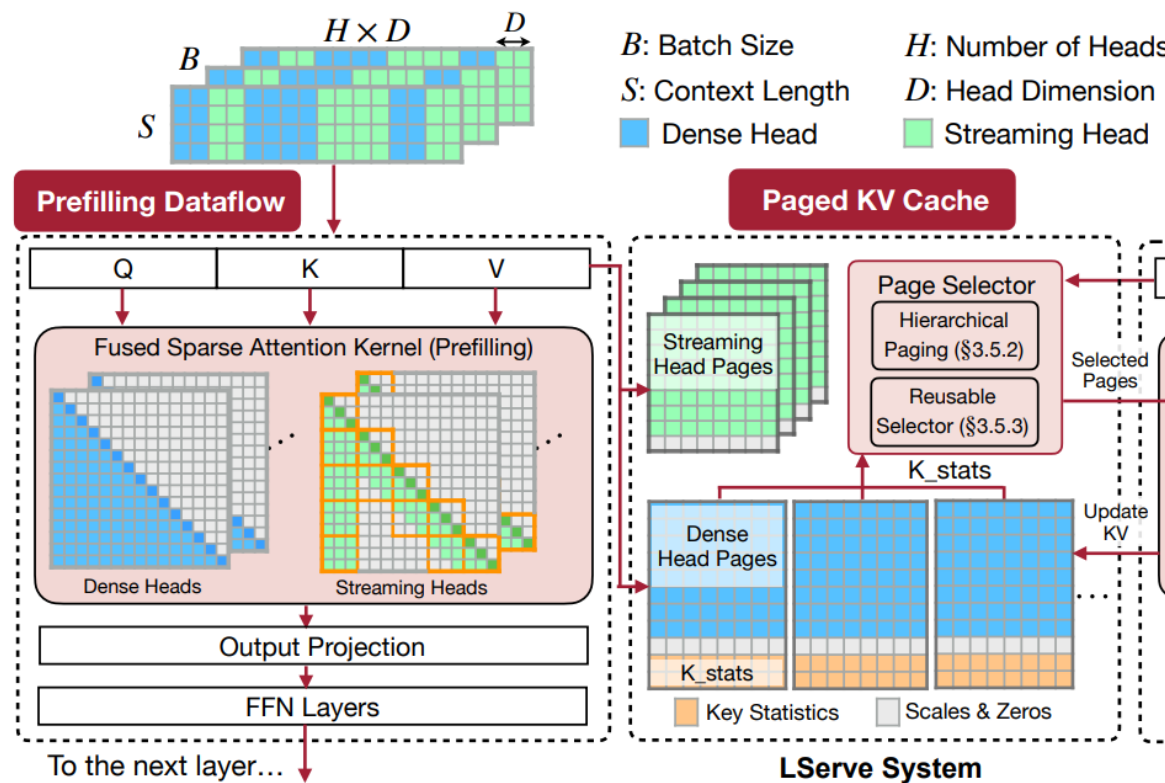
(c) Block-Sparse: Streaming Heads



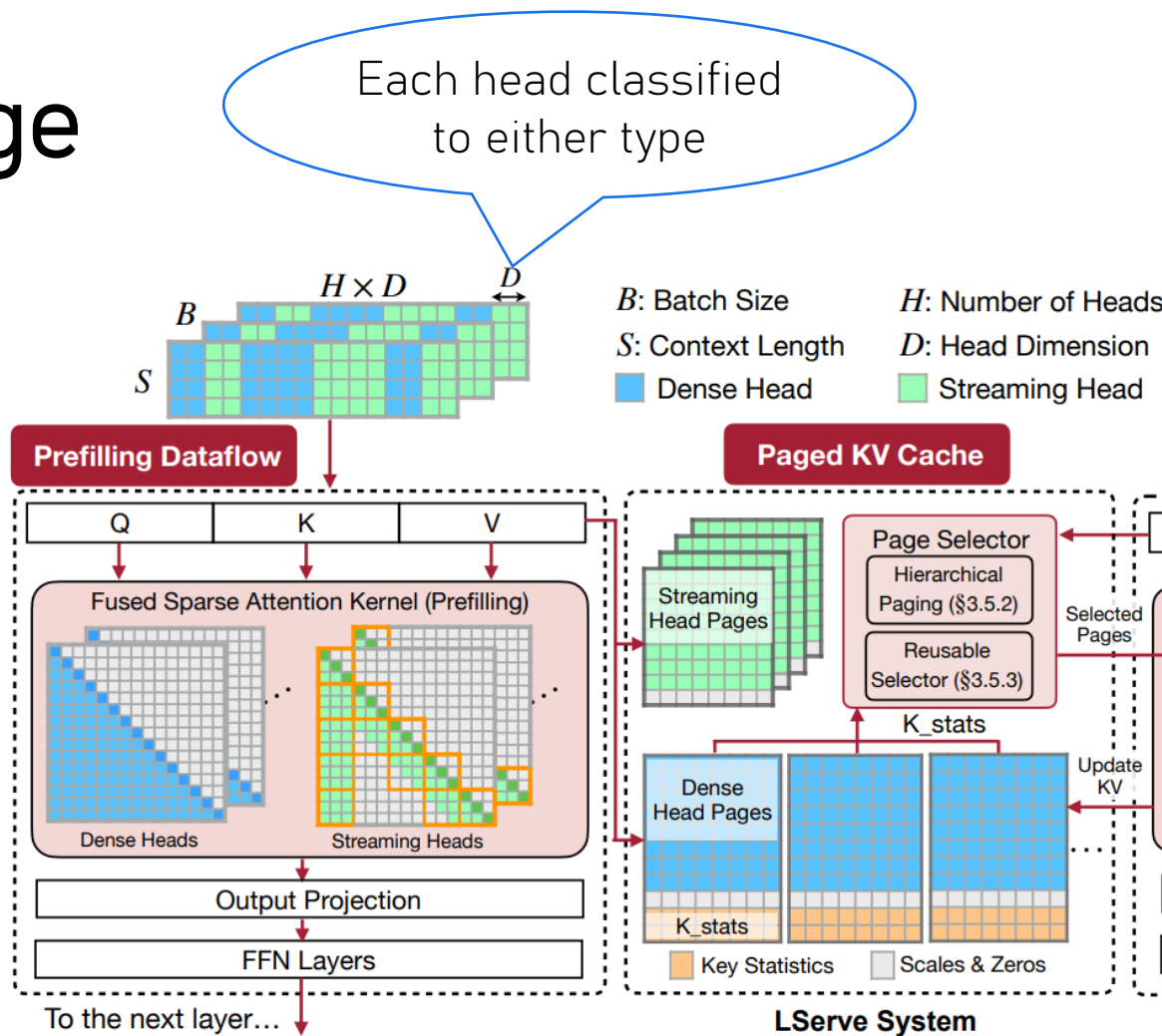
(d) Block-Sparse: Page Pruning

Dynamic, bound
the decoding
complexity to a
constant

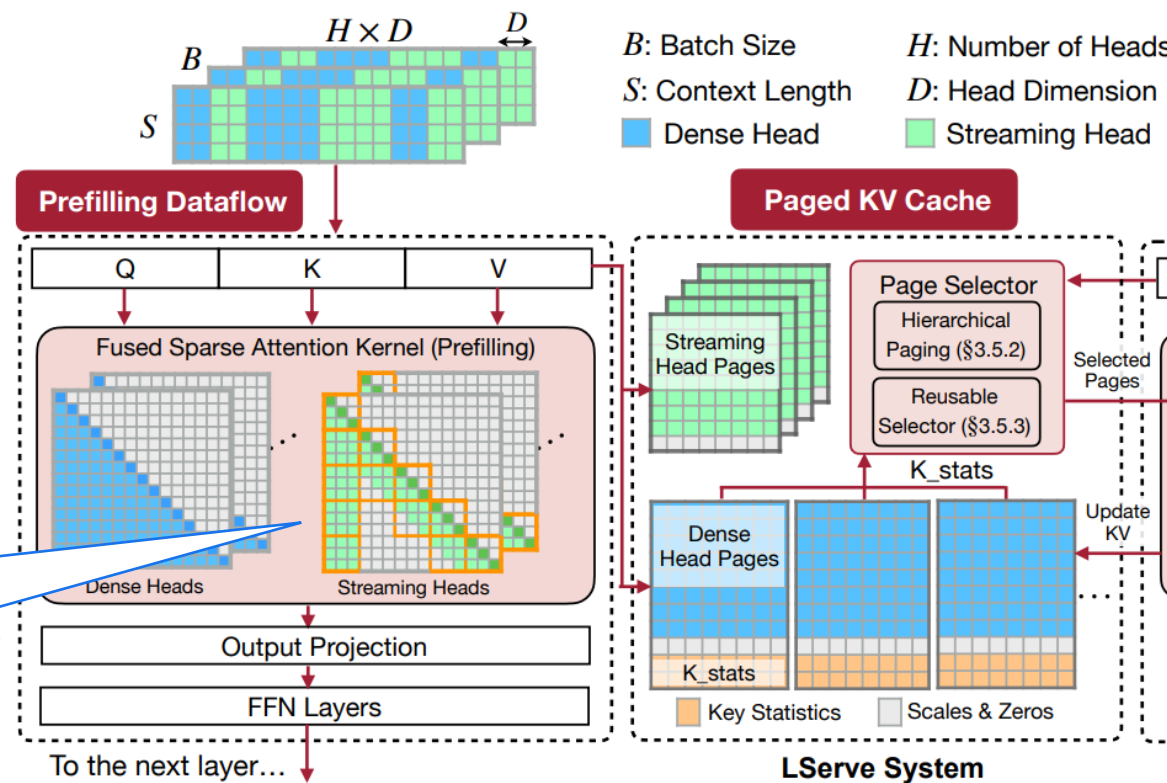
Prefill Stage



Prefill Stage

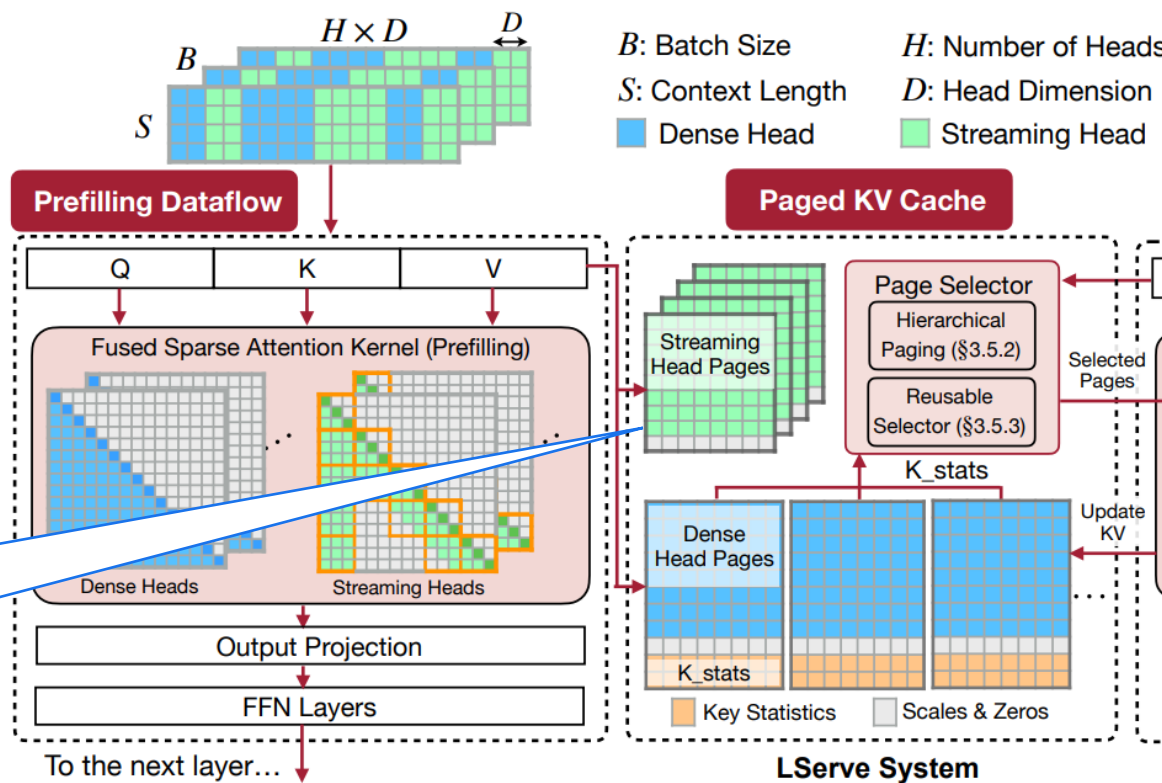


Prefill Stage



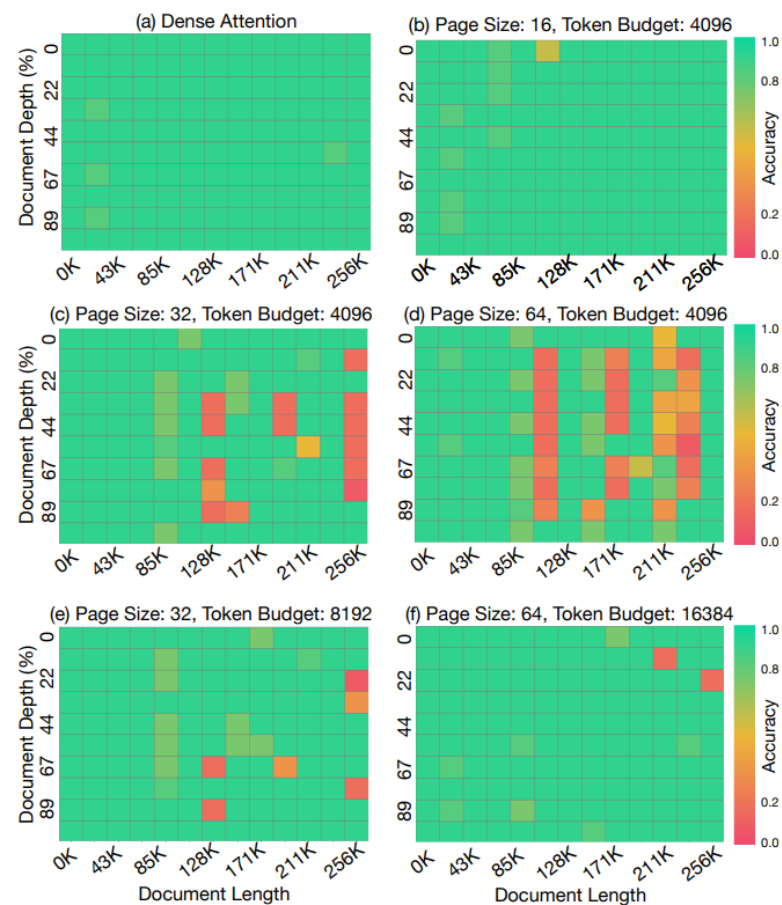
Offsets are predetermined to save computation

Prefill Stage

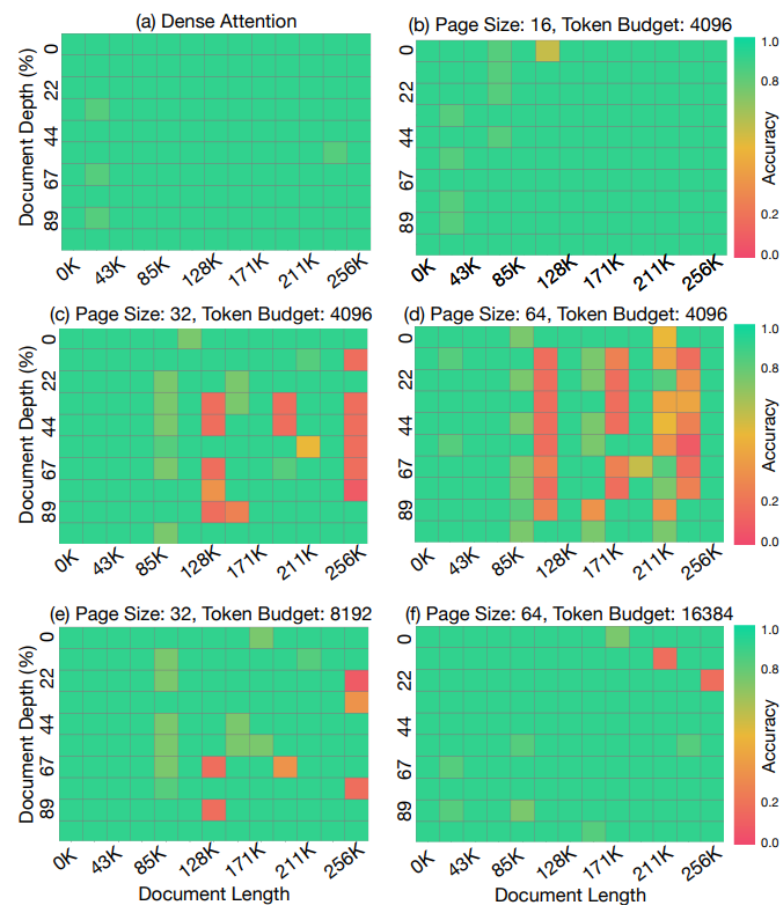


Two attention patterns are stored separately

Decoding Stage – Page Size

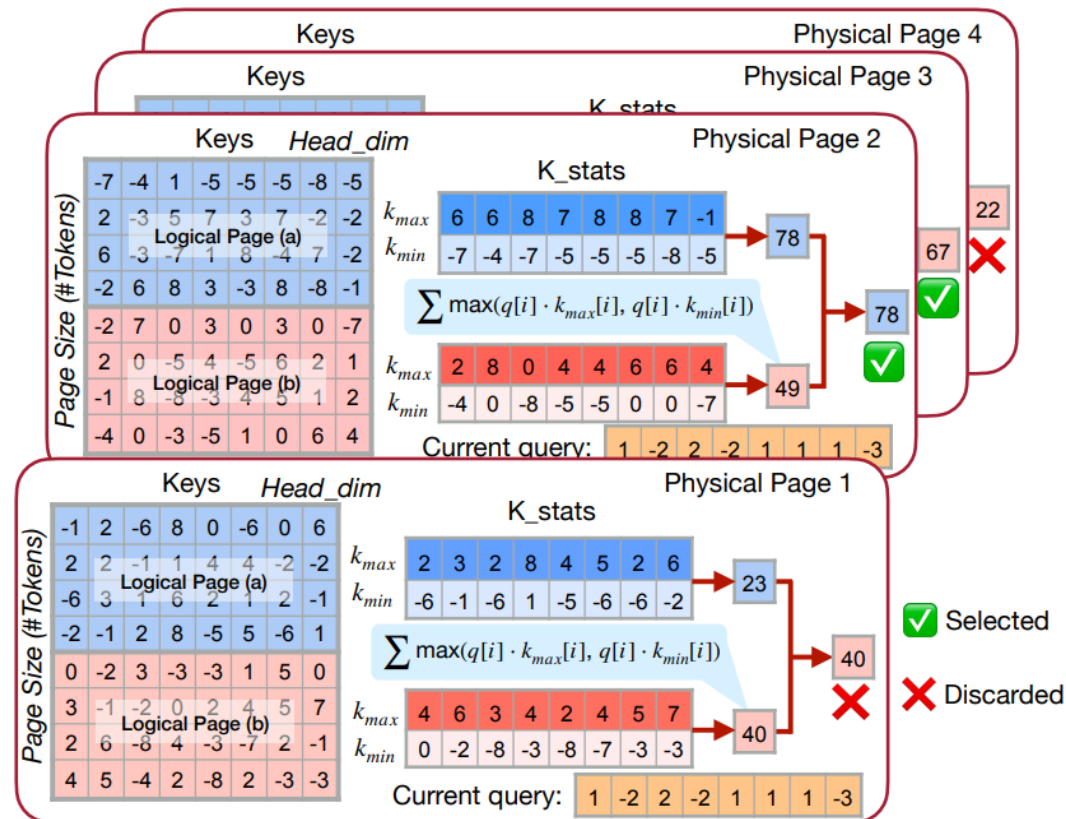


Decoding Stage – Page Size

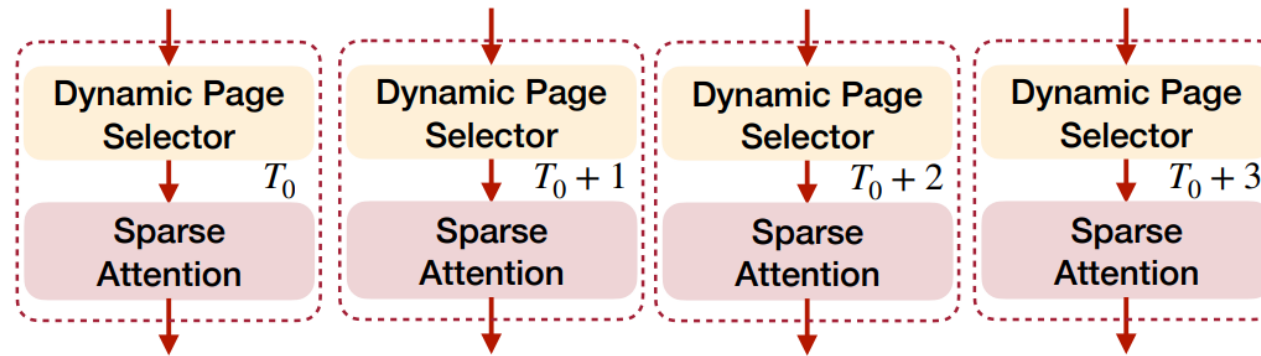


Caused by less representative page-wise statistical indicators instead of coarser granularity of sparse attention.

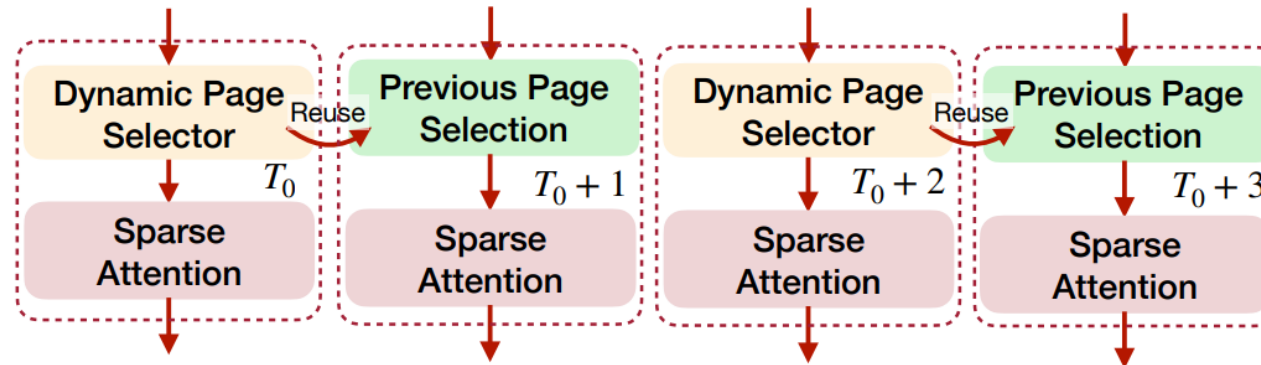
Decoding Stage – Hierarchical Paging



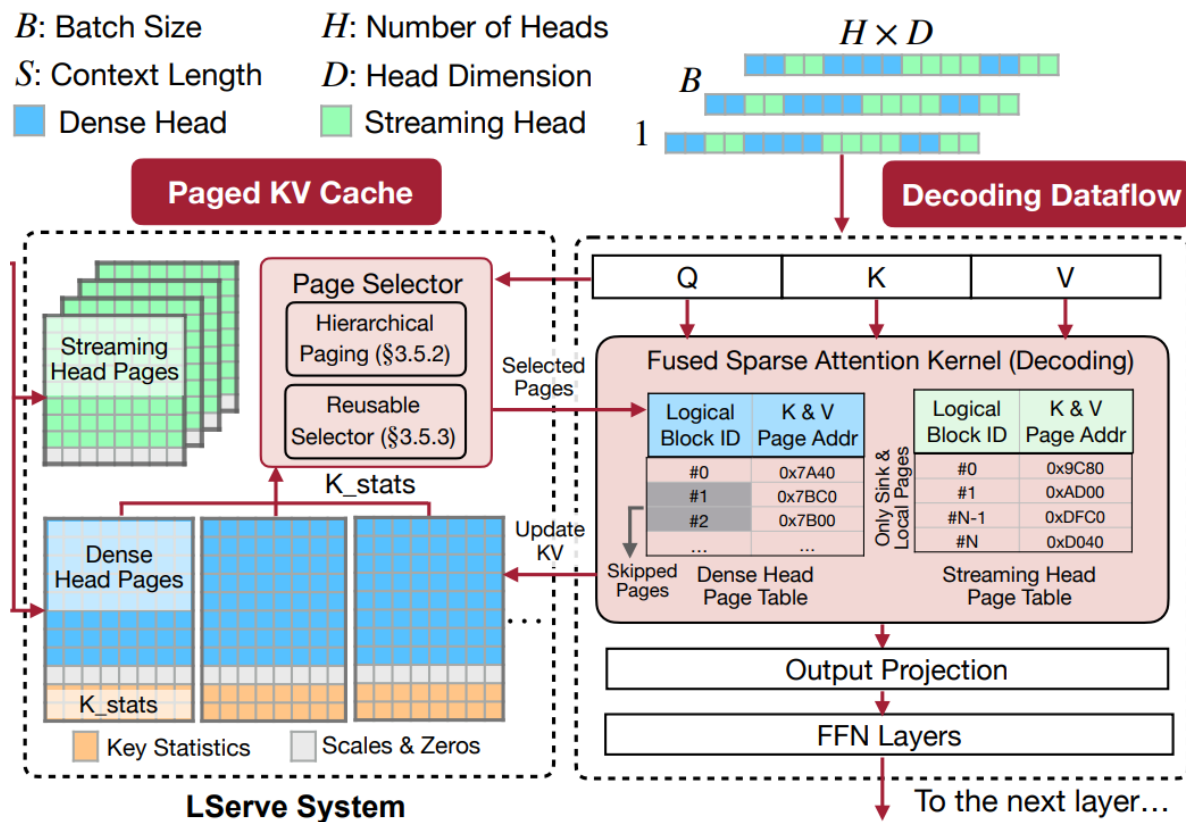
Decoding Stage – Page Selector



(a) Vanilla pipeline of dynamic sparse attention.



Decoding Stage





Evaluation – Settings

- GPUs: A100 80GB, L40S 48GB
- Models: Llama3-8B, Llama2-7B, Minitron-4B
- Metrics: throughput, TTFT, TPOT
- Baselines: vLLM, QServe, MInference, DuoAttention

Evaluation – Accuracy

Table 2: **Accuracy evaluation on LongBench** (Bai et al., 2023). We compare our method with vanilla dense attention on 2 models and 10 different benchmarks.

Model	Llama-3-8B		Llama-2-7B	
Benchmark	Dense	LServe	Dense	LServe
2WikiMQA	30.3	31.6	35.4	35.1
DuReader	30.3	30.8	25.4	24.7
HotpotQA	41.7	42.7	47.4	49.6
MultiNews	27.7	27.7	26.6	26.6
Qasper	31.7	29.3	32.6	29.5
QMSum	23.8	24.0	21.0	21.3
SamSum	41.2	39.3	41.8	41.5
TriviaQA	84.9	83.7	86.2	86.5
Average	38.9	38.6	39.5	39.4

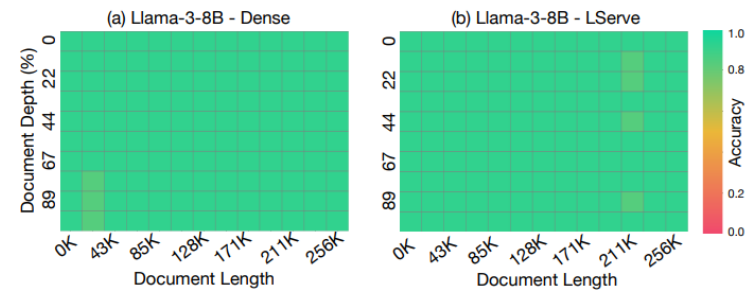


Figure 9: **Accuracy evaluation on Needle-in-a-Haystack.**

Evaluation – Efficiency

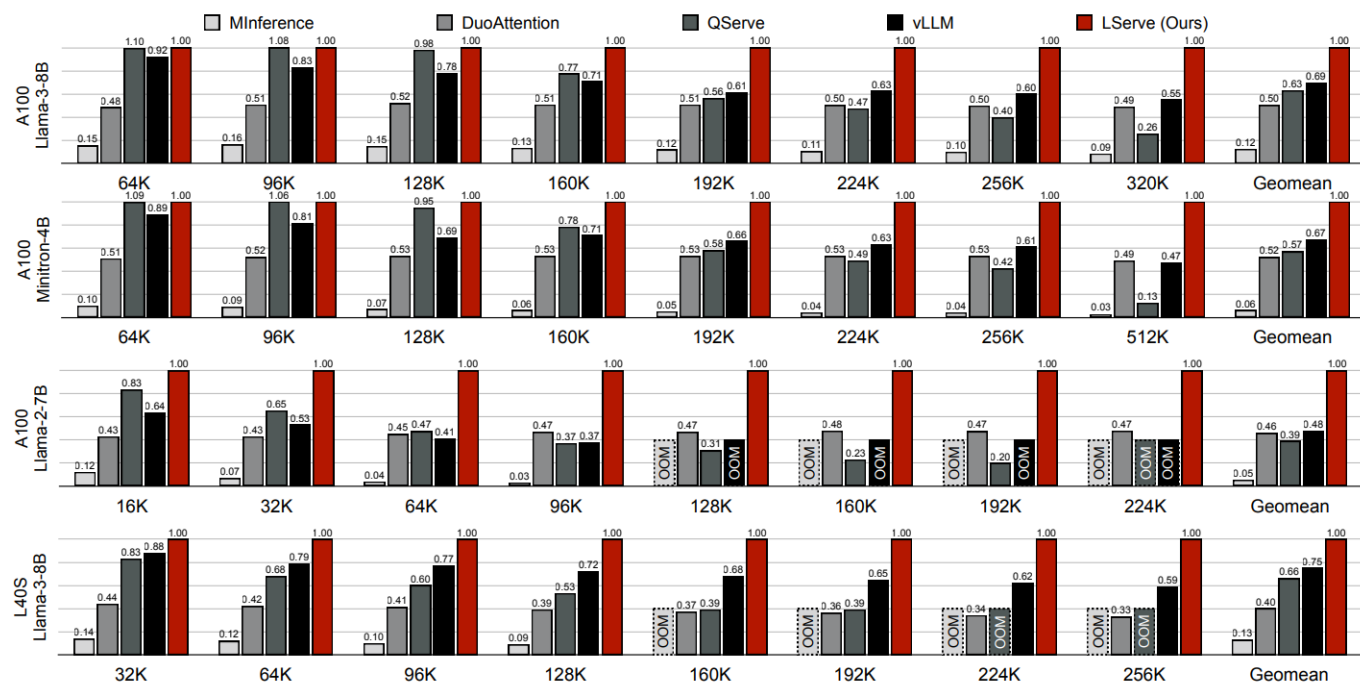


Figure 10: **Decoding Speed Evaluation.** The y-axis indicates the relative throughput of each system, normalized by the speed of LServe. Note that Minference exhibits limited decoding performance due to its unoptimized decoding stage with **dense attention**, but when integrated into vLLM, it can achieve throughput comparable to that of vLLM.

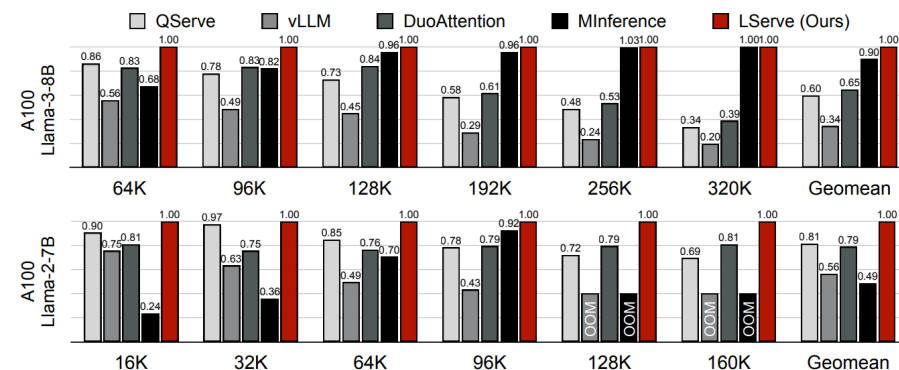
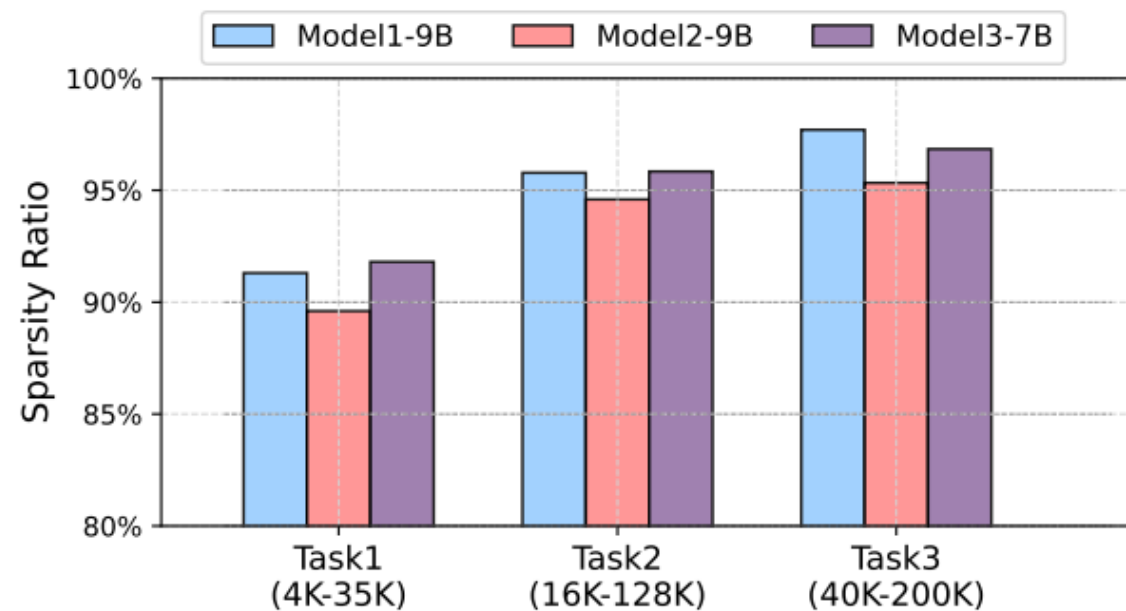


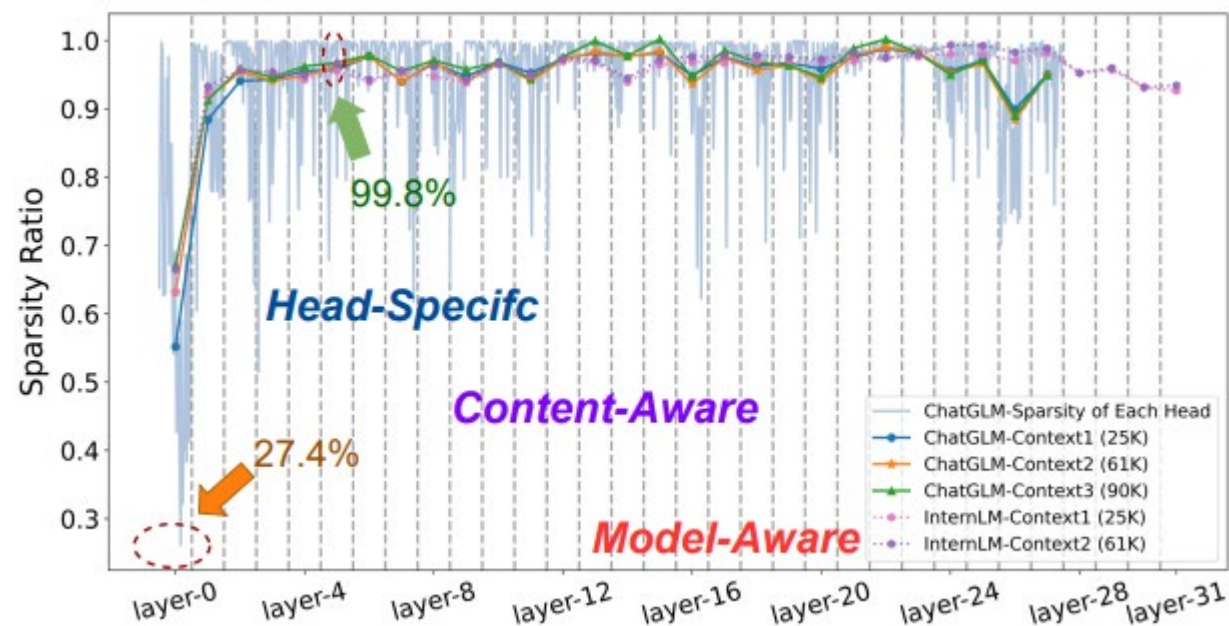
Figure 11: **Prefilling Speed Evaluation.** Performance comparison of long-sequence prefilling across different serving frameworks, normalized to LServe’s speed.

SampleAttention

Motivation – Sparsity



Motivation – Adaptive Sparsity Ratio



Motivation – Sparse Pattern

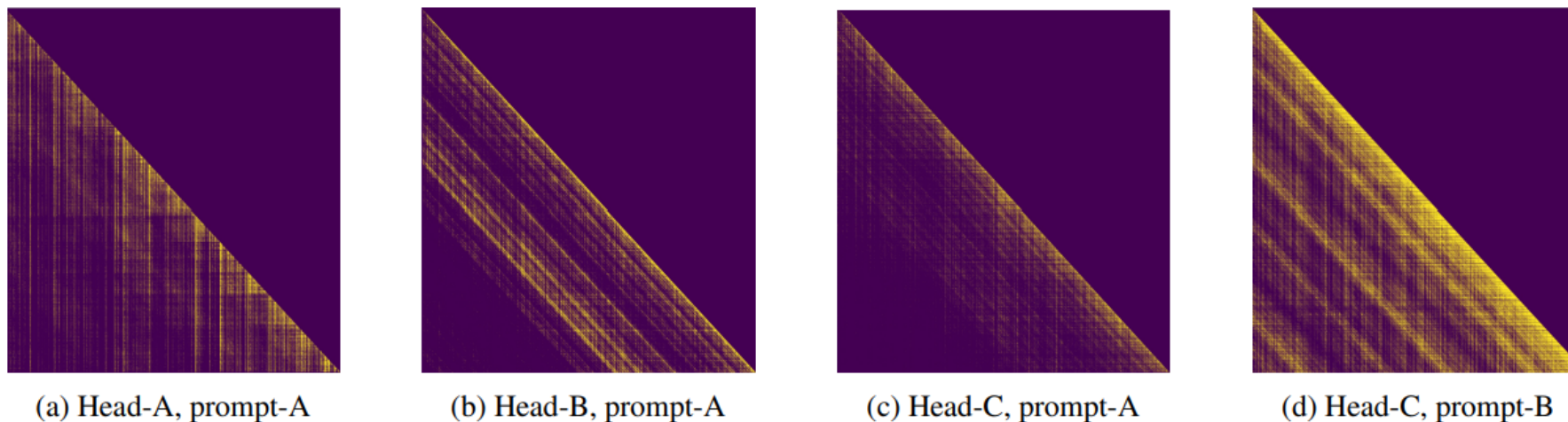
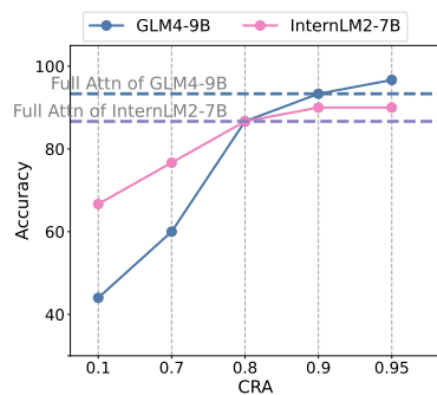
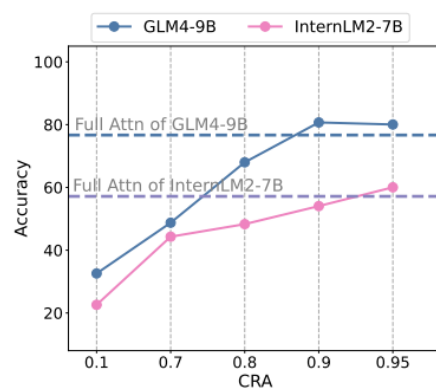


Figure 4. The visualization of attention reveals diverse structured sparse patterns. Different heads with same prompt exhibit dynamic sparse indices and ratios, but the patterns can generally be categorized into (a) **column**, (b) **slash**, or (c) **composed** pattern with column and slash. These structured patterns generally extend across the entire head. However, in some heads, such as (c), a prominent column structure in the upper half gradually fades in the lower half. Additionally, as shown in (d), the same head exhibits significant pattern differences under different prompts, highlighting the content-aware nature.

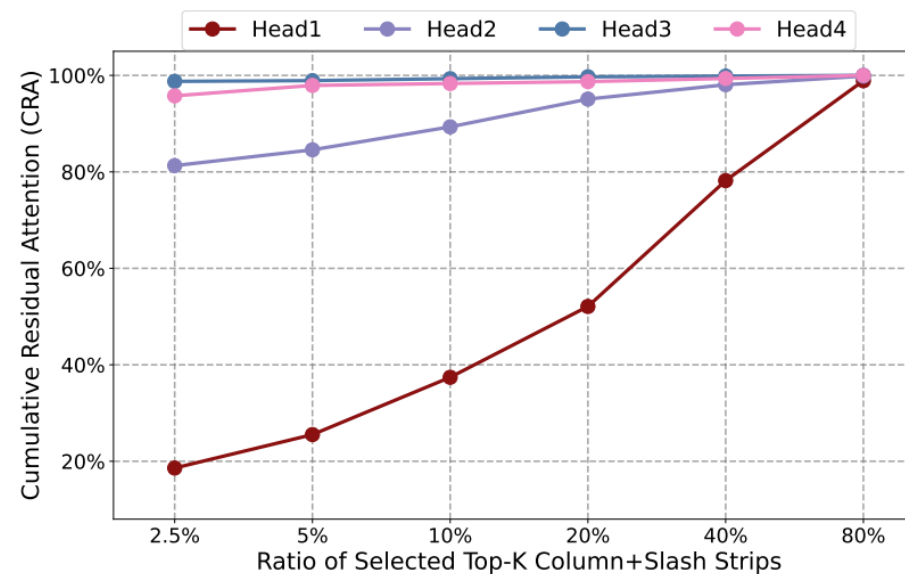
Motivation – Cumulative Residual Attention



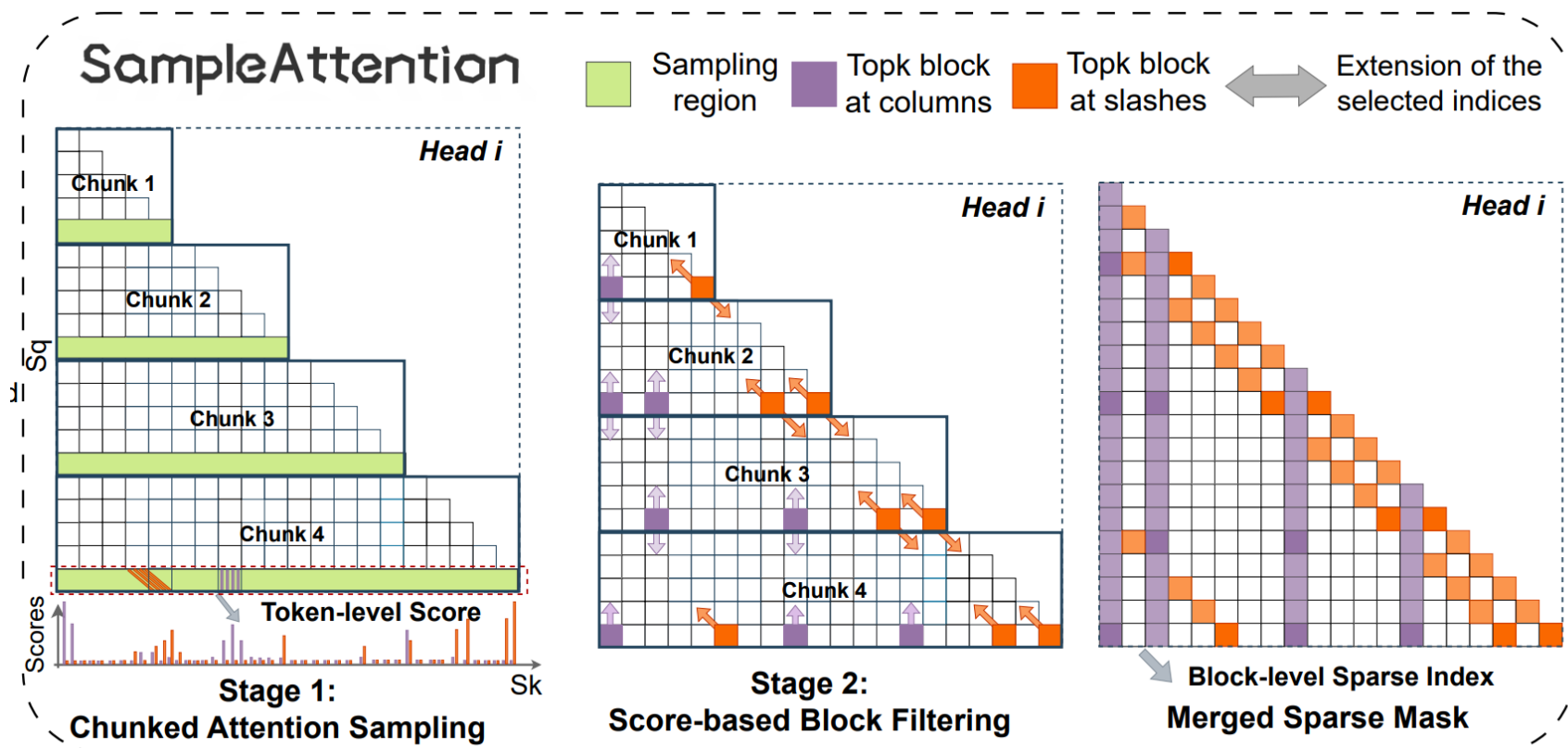
(a) Task1: Retrieval



(b) Task2: Question Answering

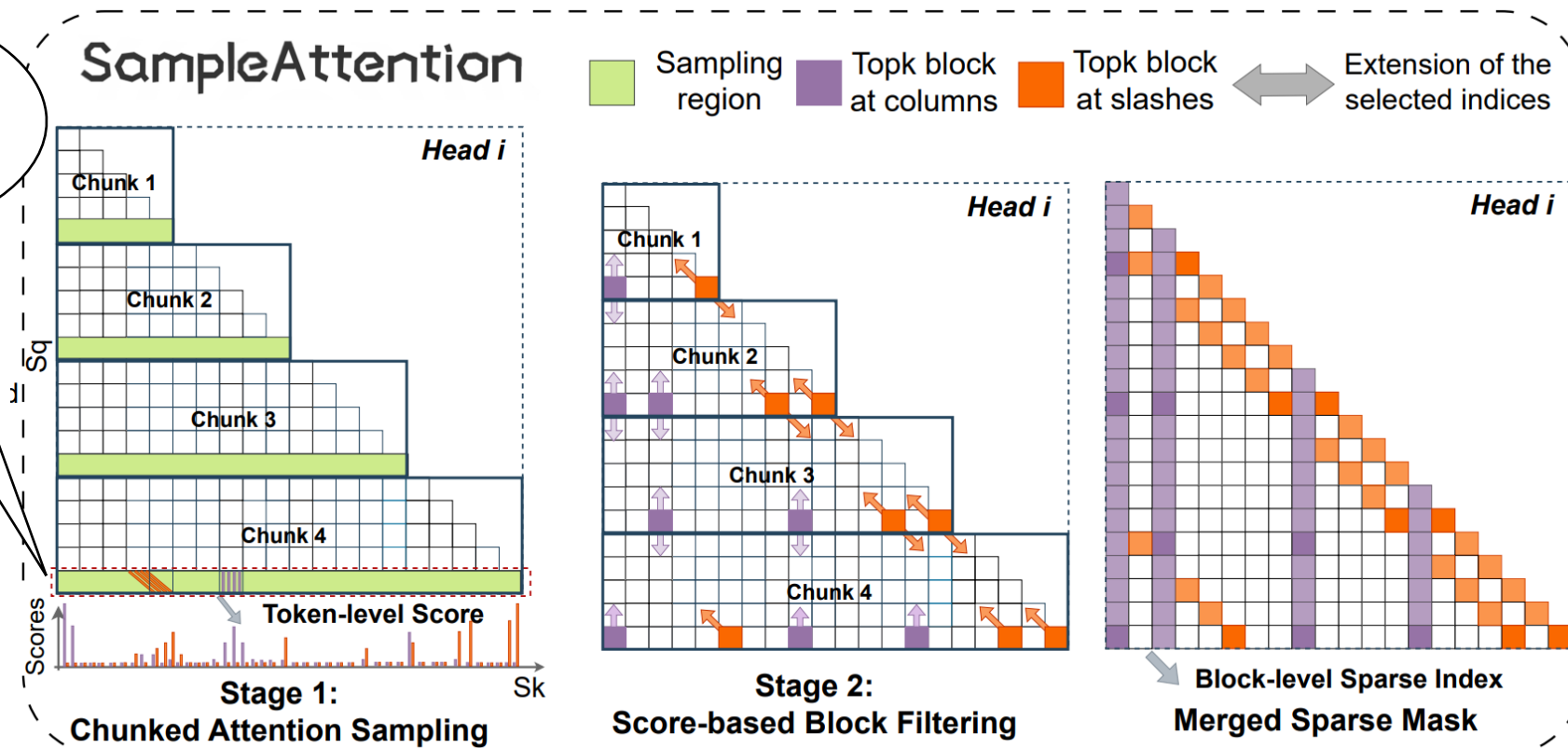


SampleAttention



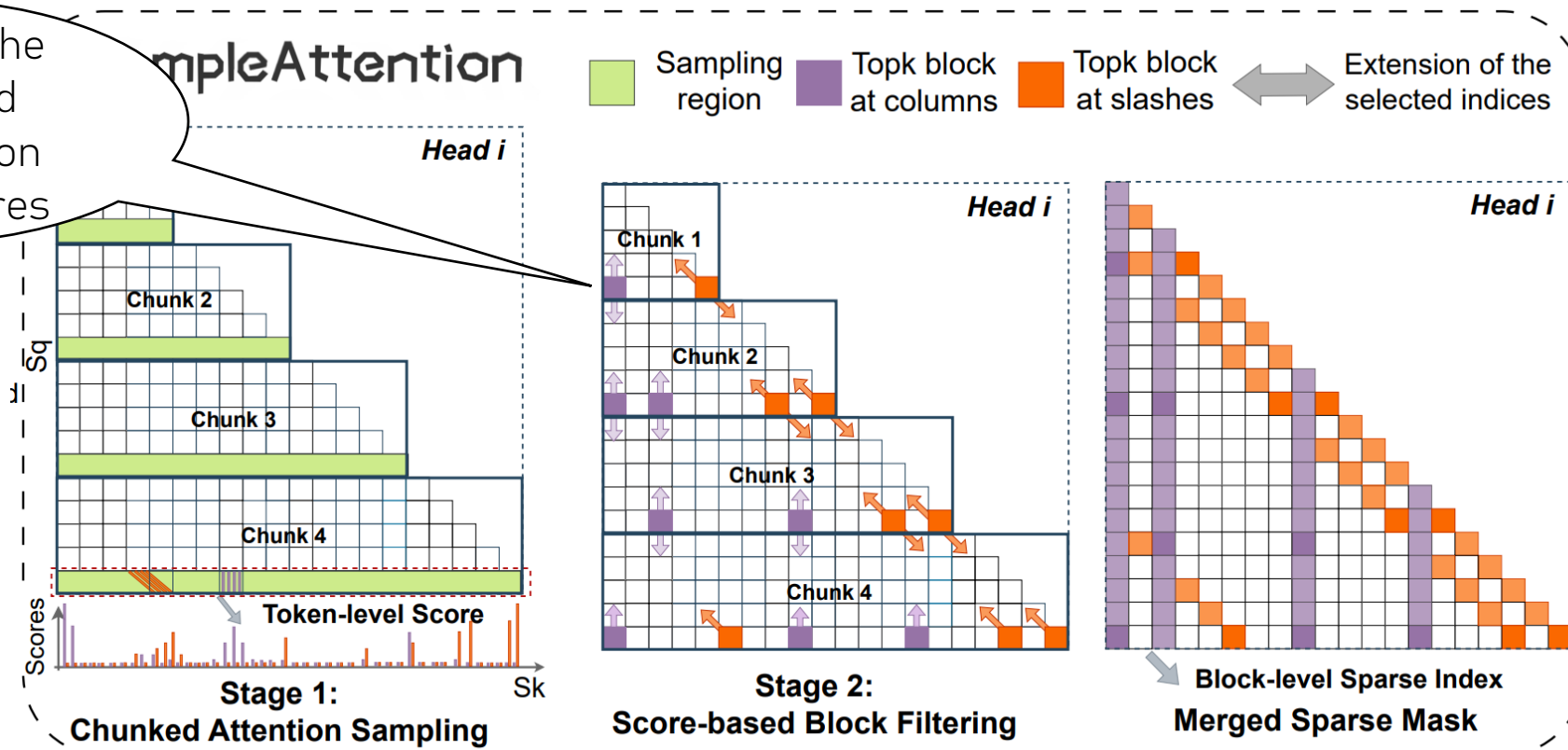
SampleAttention – Stage 1

Sample each chunk to find salient tokens

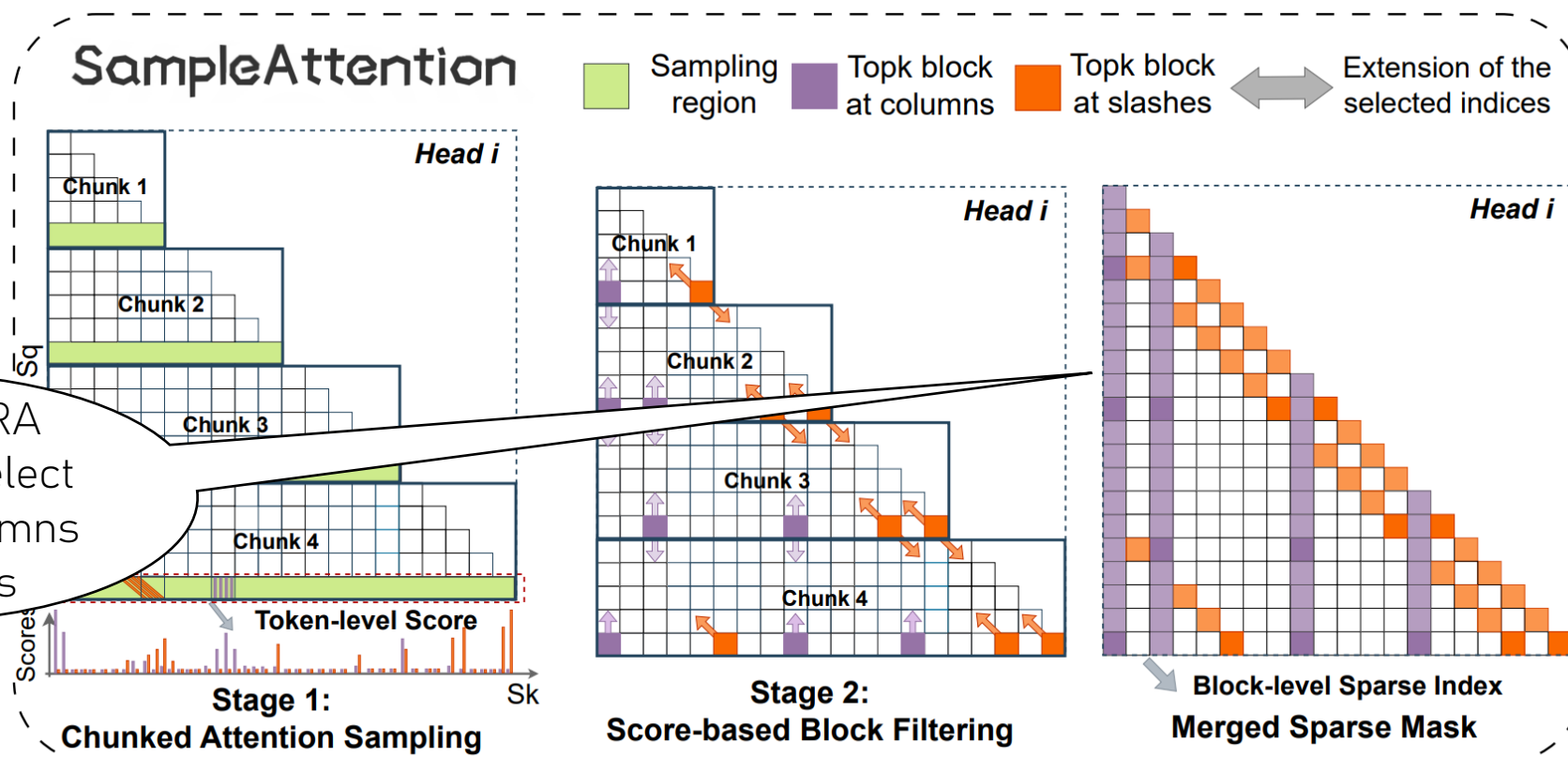


SampleAttention – Stage 2

Accumulate the column- and slash-direction attention scores



SampleAttention – Stage 2

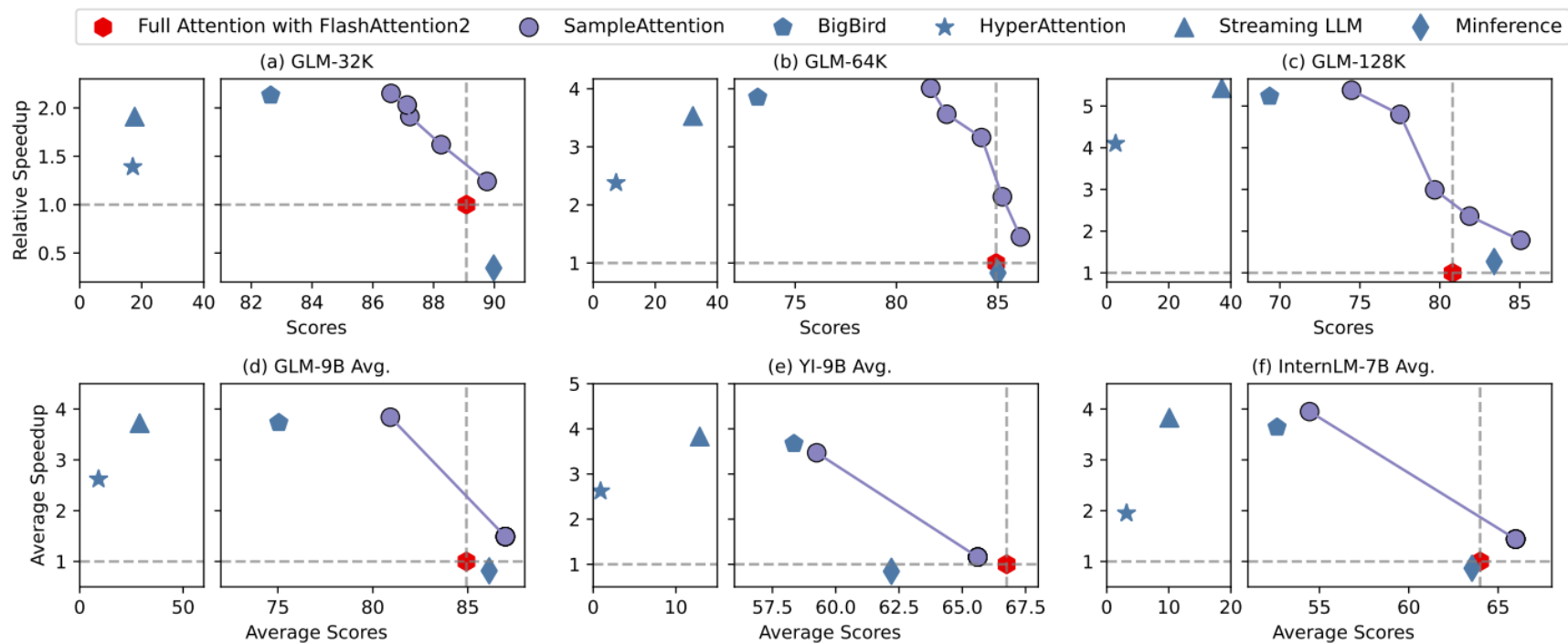




Evaluation - Settings

- Models: ChatGLM4-9B, YI-9B, InternLM2-7B
- GPU: A100 80G
- Benchmarks: RULER, LongBench, InfiniteBench
- Metrics: accuracy, speedup against FlashAttention2

Evaluation Result



Homework 1

- Read the related materials, and answer the following questions:
 - 1. Based on the paper of Llumnix, why is request rescheduling needed? What are the scenarios where request scheduling can help?
 - 2. How does Llumnix mitigate the time latency caused by live migration?
 - 3. How does KunServe serve requests efficiently after dropping the parameters?
- Related materials:
 - Llumnix paper: <https://www.usenix.org/system/files/osdi24-sun-biao.pdf>
 - KunServe paper: https://ipads.se.sjtu.edu.cn/_media/publications/kunserve-eurosys26.pdf

Homework 2

- Write a paper review for [*DEEPSERVE: Serverless Large Language Model Serving at Scale*](#), following the format below:
- **Overall score** (1: reject, 2: weak reject, 3: major revision, 4: minor revision, 5: accept)
- **Summary of the paper** (be concise, in one paragraph)
- **Reasons for accepting the paper** (list in a pointed format, i.e., "- reason1, - reason 2")
- **Reasons for rejecting the paper** (the same as the reasons for acceptance)
- **Comments for the authors** (the detailed opinions on this paper, including explaining your score, asking questions, suggesting improvements, etc.)



Paper Review Format

- Overall score
 - Reject: serious concerns
 - Weak reject: concerns but may not be strong enough
 - Major revision: need to add experiments, designs, etc.
 - Minor revision: writing issues, etc.
 - Accept: almost no need to improve



Paper Review Format

- Overall score
- Summary
 - Be concise
- Reasons
 - Using a pointed format:
 - - Reason 1
 - - Reason 2



Paper Review Format

- Overall score
- Summary
- Reasons
- Comments
 - Be polite
 - Give details about your reasons
 - Questions, necessary improvements (especially when you give a major/minor revision score)

Q & A

